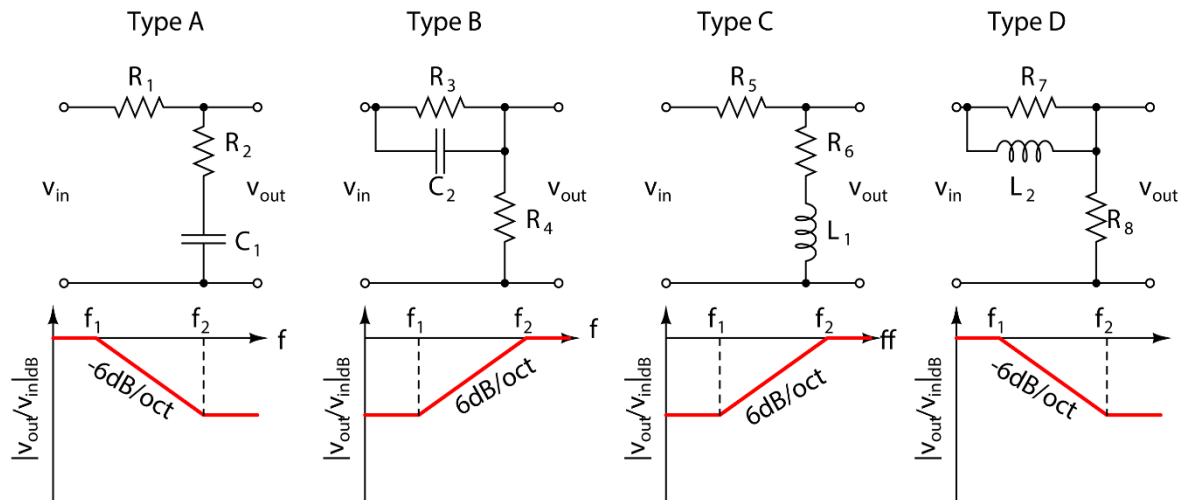


EE202 LAB #4

FREQUENCY RESPONSE OF RC and RL CIRCUITS

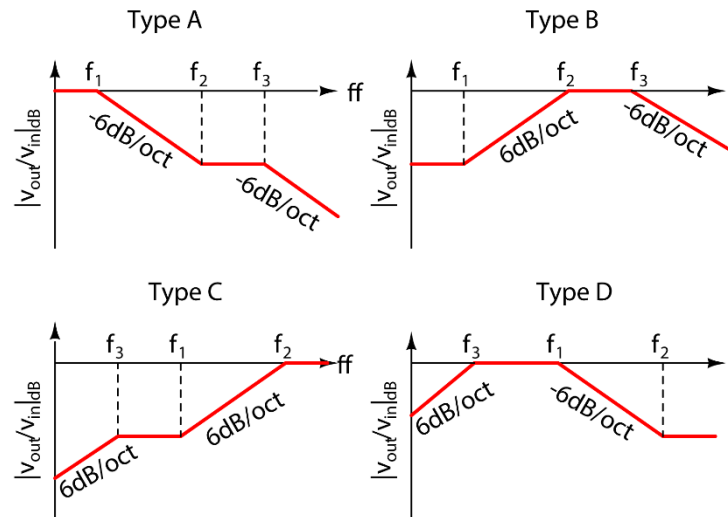
Preliminary Work:

1. This part aims to design a first-order RC or RL circuit with the given frequency response.



The corner frequencies (f_1 and f_2) and type of the circuit are assigned to you.

2. Derive the transfer function for your circuit. Plot the theoretically expected transfer function using MATLAB $|V_{out}/V_{in}|_{dB}$ on a logarithmic frequency axis. You can use `semilogx` function in MATLAB for the plot.
3. Determine corner frequencies in terms of circuit parameters. Determine the component values using standard capacitors and resistors. Do not choose any capacitor smaller than 100pF. Do not choose any resistor larger than 500K Ω . For type C and D circuits, use the 15 μ H hi-Q inductor available in the lab.
4. Perform an LTSpice AC analysis of the designed circuit. Check that your circuit satisfies the desired characteristics.
5. Add another corner frequency (**which is assigned to you**) to your circuit using an RC circuit to generate the required characteristics as shown. Since f_3 is sufficiently far away, the other two corner frequencies should stay nearly the same.



- Derive the transfer function for the new circuit.
- Plot the transfer function using LTSpice AC analysis. Check that the circuit behaves as expected.

Experimental Work:

First-order RC/RL circuit

Type	f_1 (KHz)	f_2 (KHz)

Build your RC/RL circuit on your breadboard using the values you found. Apply the signal generator at the input. Set the amplitude to 5Vpp. Fill the table

f	Freq (KHz)	v_{out} (v)	v_{out}/v_{in}	v_{out}/v_{in} (dB)	f	Freq (KHz)	v_{out} (v)	v_{out}/v_{in}	v_{out}/v_{in} (dB)
$0.1f_1$					$0.4f_2$				
$0.2f_1$					$0.8f_2$				
$0.4f_1$					f_2				
$0.8f_1$					$1.2f_2$				
$1.0f_1$					$2f_2$				
$1.2f_1$					$4f_2$				
$2f_1$					$8f_2$				

Show that your RC/RL circuit has $\pm 6\text{dB/octave}$ in two regions.

Show your measurement results to the assistant to get a check.

For your report, plot the dB gain on a semilog frequency axis (use MATLAB) to be compared with the LTSpice results.

Return the inductor to the lab technician when the measurements are finished.

Second-order RC/RL circuit:

Using the designed values, build your second-order RC/RL circuit on the breadboard.

Measure the transfer function of the filter circuit, $v_{\text{out}}/v_{\text{in}}$, where v_{out} is the peak-to-peak value at the output and v_{in} is twice the pp value shown on the signal generator screen. Use EXT SYNC facility of the oscilloscope for problem-free measurements. Measure the frequency range from 0.1 times the smallest corner frequency to 10 times the highest corner frequency. Choose the frequencies in the table accordingly.

f_3 (KHz)	
-------------	--

f_0	Freq (KHz)	v_{out} (V)	$v_{\text{out}}/v_{\text{in}}$	$v_{\text{out}}/v_{\text{in}}$ (dB)	f_0	Freq (KHz)	v_{out} (V)	$v_{\text{out}}/v_{\text{in}}$	$v_{\text{out}}/v_{\text{in}}$ (dB)

Compare the filter's gain at the center frequency with the intended value.

Show that the filter has an asymptotic response of 6dB/octave at low frequencies, and -6dB/octave at high frequencies.

Show the measurement results to the assistant to get a check.

For your report, plot the measured dB results on a semilog frequency axis (use MATLAB) to compare with LTSpice results.